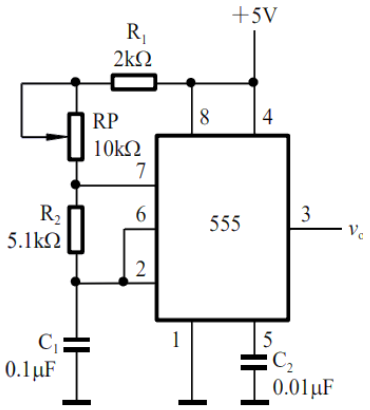
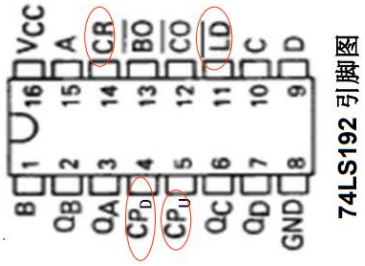
## 信号源（振荡器）



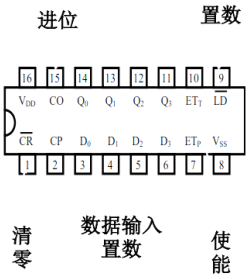

具体参数：（100Hz~100 分频）

## 计数器（异步清零）

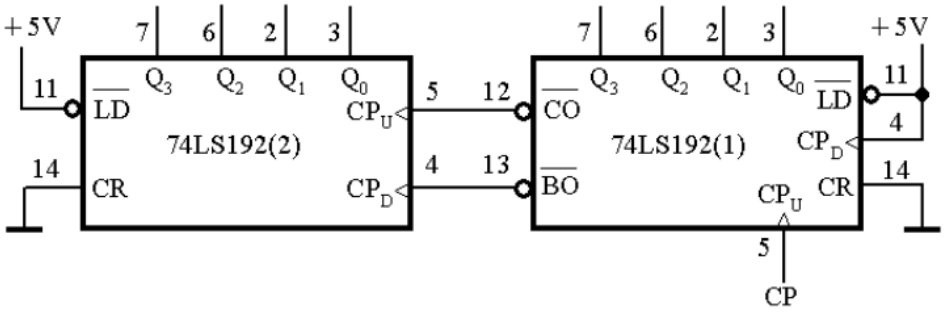
74HC192（异步置数）（半个周期 CO）



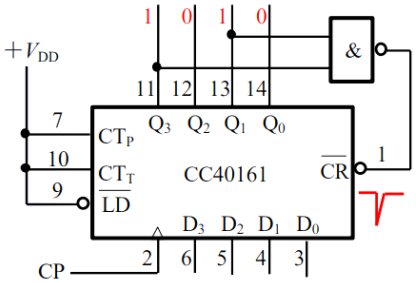

74HC161（同步置数）（一个周期 CO）



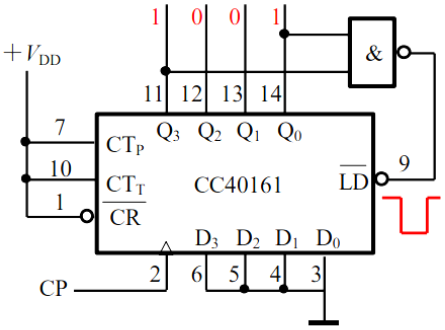

## 可加减（100）



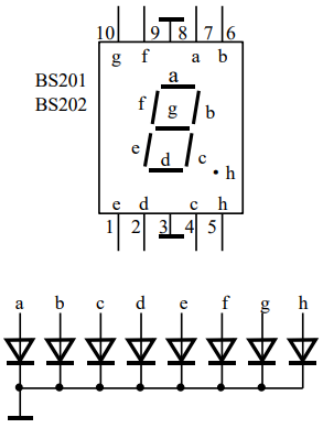
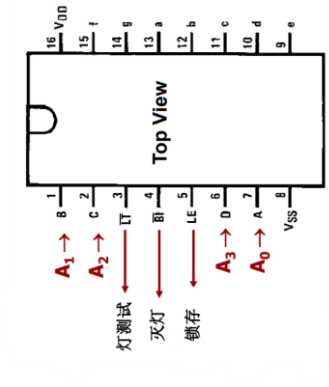
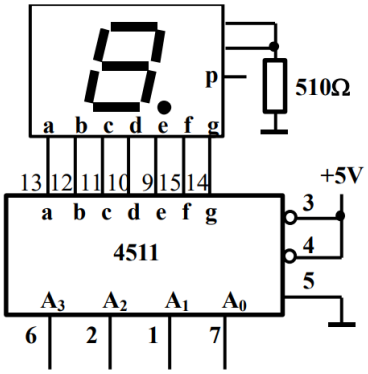
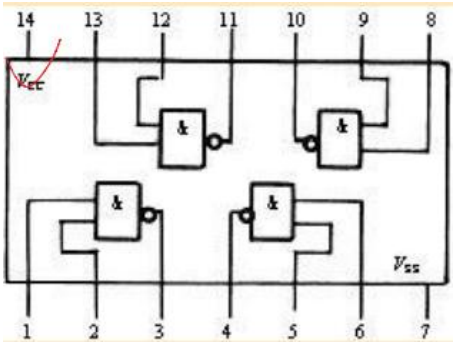
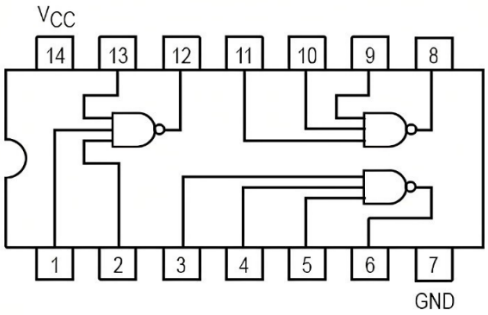

## 加（10,0~9）



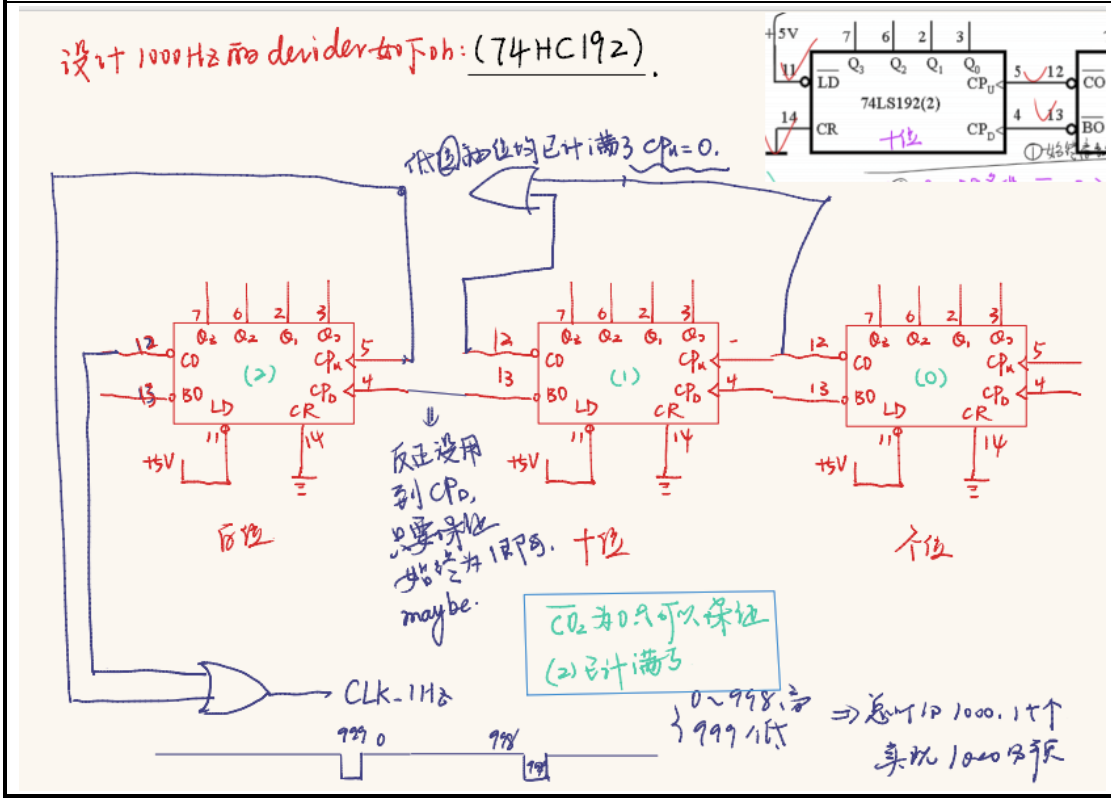

利用异步清零



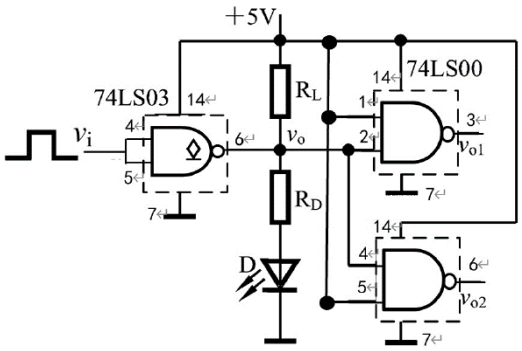

利用同步预置→清零

| 译码电路 | |
|---|---|
| 共阴极数码管 | CD4511（**Vss** 为负电源） |
|  |  <br>（0~9） |
|  | |
| 其他元件 | |
| CD4511（与非门） | 74hc10（三输入） |
|  |  |
| | |
| | |

## 分频器（1000）




## OC 门




公式：

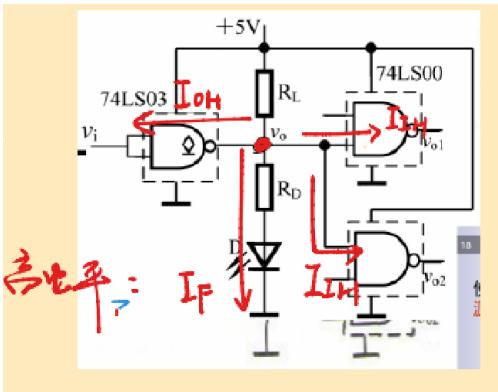
1. $R_D = \dfrac{V_O - V_F}{I_F} = 1.75k\Omega$，取 2kΩ

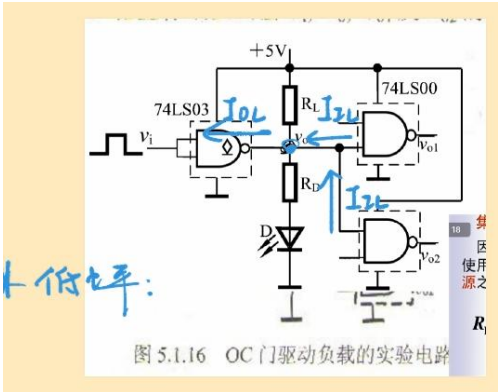
2. 参数：

   a) 74HC03：VOHmin=2.4V，VOLmax=0.4，IOHmax=0.5uA，IOLmax=5.2mA。

   b) 74HC00：IIHmax=1uA，IILmax=1uA。

   c) VF=1.5V，IF=2mA。

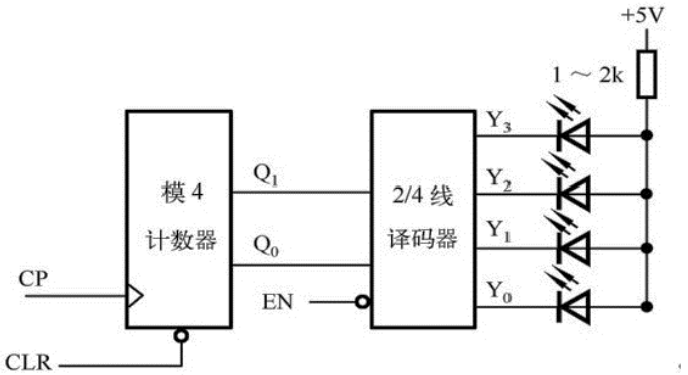


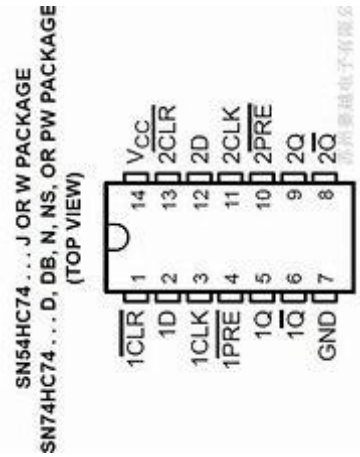

$$R_{L\max} = \frac{V_{CC} - V_{OH\min}}{I_{OH} + I_F + 2I_{IH}} = 1298.4\Omega$$

$$R_{L\min} = \frac{V_{CC} - V_{OL\max}}{I_{OL} - 2I_{IL}} = 885.0\Omega$$

流水灯电路（各个 LED 灯依次点亮）




74hc74（D trigger）




2-4 译码逻辑

| $\overline{EN}$ | $Q_1$ | $Q_0$ | $\overline{Y_3}$ | $\overline{Y_2}$ | $\overline{Y_1}$ | $\overline{Y_0}$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | X | X | 1 | 1 | 1 | 1 |

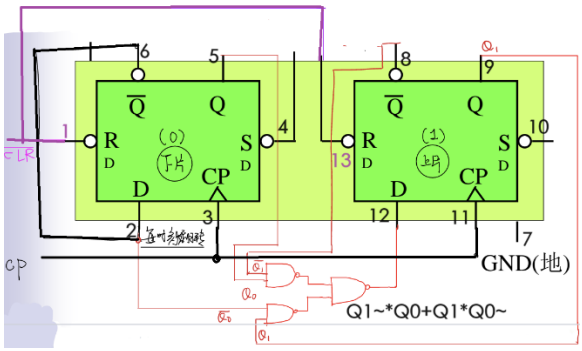
$\overline{Y3}=\overline{ENQ1Q0}$

$\overline{Y2}=\overline{ENQ1\overline{Q0}}$

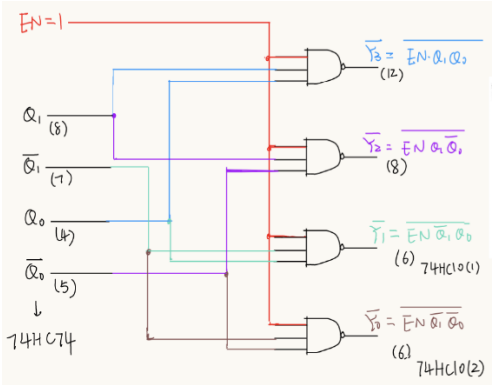
$\overline{Y1}=\overline{EN\overline{Q1}Q0}$

$\overline{Y0}=\overline{EN\overline{Q1}\,\overline{Q0}}$

4 进制计数器（引脚也许有问题）




2-4 译码器

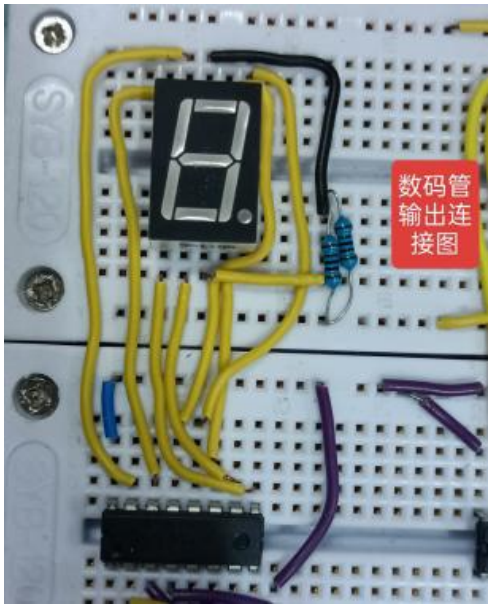
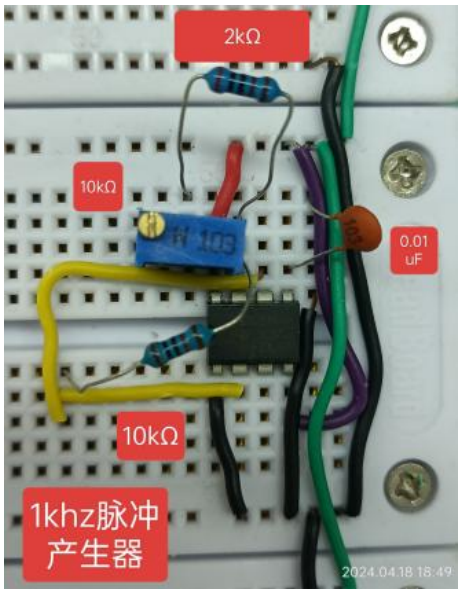
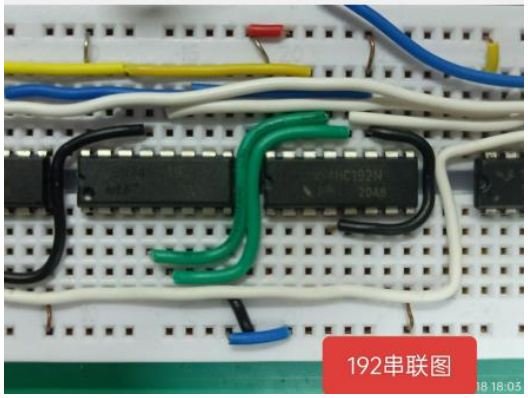



$Q_0$ 每次反转，以及 $Q_1^{n+1}$ 等于 $Q_0$^$Q_1$

让 EN=1（而不是对～EN 取非）

观察波形

| 加计数：下降沿 | 减计数：上升沿 |
|---|---|
| 触发信号：最高位（保持不变） | 触发方式：边沿触发 |

| 实图 | |
|---|---|
|  数码管输出连接图 |  2kΩ<br>10kΩ<br>0.01 uF<br>10kΩ<br>1khz脉冲产生器<br>2024.04.18 18:49 |
|  192串联图 | |
| | |